Flow Control at Data Link Layer using sliding window protocol

Exp.no : 7

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Aim:

To write a program to implement flow control at the Data Link Layer using the sliding window protocol, and to simulate the flow of frames from one node to another.

Algorithm:

1. Start the program
2. Read the total number of frames to be sent (N), the sender's window size (W) and (optionally) a frame number to simulate as lost during transmission.
3. Initialize base = 0 (sequence number of the oldest unacknowledged frame) and nextseq = 0 (sequence no. of the next frame to be sent).
4. Repeat while base < N:
   a) Determine the current sender window as the range [base, base + W - 1].
   b) Send all frames with sequence numbers from nextseq up to (base + W - 1), incrementing nextseq after each frame, since the window restricts the sender to at most W outstanding (unacknowledged) frames at any time.

c.) If the lost frame lines within the frames just sent :-

(i) The receiver discards the lost frame and any subsequent out-of-order frames.

(ii) The sender times out and retransmits all frames starting from the the lost frame (Go-Back-N)

d.) Else (no loss in this batch):

(i) The receiver accepts each frame in order and sends a cumulative acknowledgement (ACK) for it.

(ii) On receiving the ACK for the last frame sent, the sender slides its window forward :

base = nextSeq.

5.) Repeat step 4 until all N frames have been transmitted and acknowledged

6.) Stop the program.

program

```
def simulate (total_frames, window_size, lost_frame):
    base = 0
    next_seq = 0
    while base < total_frames :
        window_end = min( base + window_size - 1, total_frames
        print ("\n----------")
        print (f"Sender window : [Frame {base} - Frame {window-
                                                    end}]")
        while next_seq < base + window_size and next_seq <
        total_frames :
            print (f"Node A --> Sending Frame {next_seq}")
            if next_seq == lost_frame :
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        print (f" : (Frame {next_seq} Lost in transit!)")
    next_seq += 1

    if lost_frame is not None and base <= lost_frame
    < next_seq:
        print (f"Node B -> Frame {lost_frame} missing.
            Discarding out-of order frames after it.")
        print (f"Node A --> Time out! Resending window starting
            from Frame {lost_frame}.")
        next_seq = lost_frame
        lost_frame = None
        for f in range (base, next_seq):
            print (f"Node B --> Received Frame {f}, sending Ack
                {f}")
        print {f"Node A --> Ack {next_seq-1} received. Sliding
            window forward.")
        base = next_seq
    print ("\n----------------")
    print (f"All {total_frames} frames successfully
        transmitted from Node A to Node B.")

def main():
    print ("=== Sliding window protocol - Flow control
        simulation (Go-Back-N) ===\n")
    total_frames = int (input ("Enter total no. of frames
        to send :"))
    window_size = int (input ("Enter Sender's window size:"))
    lost_frame = int (input (f" Enter frame number to
        simulate as LOST \n (0 to {total_frame-1},
```

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enter -1 for no loss") : "))

if lost_frame == -1 :

lost_frame = None

```
=== SLIDING WINDOW PROTOCOL - FLOW CONTROL SIMULATION (Go-Back-N) ===

Enter total number of frames to send     : 8
Enter sender's window size               : 4
Enter frame number to simulate as LOST
(0 to 7, enter -1 for no loss)           : 5

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Sender Window : [Frame 0 - Frame 3]
Node A --> Sending Frame 0
Node A --> Sending Frame 1
Node A --> Sending Frame 2
Node A --> Sending Frame 3
Node B --> Received Frame 0, sending ACK 0
Node B --> Received Frame 1, sending ACK 1
Node B --> Received Frame 2, sending ACK 2
Node B --> Received Frame 3, sending ACK 3
Node A --> ACK 3 received. Sliding window forward.

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Sender Window : [Frame 4 - Frame 7]
Node A --> Sending Frame 4
Node A --> Sending Frame 5
        (Frame 5 LOST in transit!)
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Frame 5 missing. Discarding out-of-order frames after it.
Node A --> Timeout! Resending window starting from Frame 5.

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Sender Window : [Frame 4 - Frame 7]
Node A --> Sending Frame 5
Node A --> Sending Frame 6
Node A --> Sending Frame 7
Node B --> Received Frame 4, sending ACK 4
Node B --> Received Frame 5, sending ACK 5
Node B --> Received Frame 6, sending ACK 6
Node B --> Received Frame 7, sending ACK 7
Node A --> ACK 7 received. Sliding window forward.

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All 8 frames successfully transmitted from Node A to Node B.
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Result :

The program to implement flow control at the Data Link Layer using the Sliding window protocol was successfully written in Python, executed and verified by simulating the flow of frames b/w two node

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